Pipeline traces

These are actual Logisim-Evolution samples for assignment Sections 8.2 to 8.4. Each row is the settled state before the next rising edge. Addresses and data are hexadecimal; cycle numbers are decimal.

ALU forwarding (no unnecessary stall)

Initial register values:

R1 = 05, R2 = 03, R3 = 0C (hexadecimal).

Cycle-by-cycle pipeline trace

Cycle	IF PC	ID	EX	MEM	WB	Stall	Flush	Fwd A/B	Writes at edge
1	00	--	--	--	--	0	0	-/-	-
2	01	00	--	--	--	0	0	-/-	-
3	02	01	00	--	--	0	0	-/-	-
4	03	02	01	00	--	0	0	RF/EM	-
5	04	03	02	01	00	0	0	-/-	R1=08
6	05	04	03	02	01	0	0	-/-	R3=04

In cycle 4, SUB R3,R1 is in EX while ADD R1,R2 is in MEM (EX/MEM pipeline register).

The ADD result (08) is forwarded to operand B of SUB through the EX/MEM forwarding path.

Therefore, SUB uses the updated R1 value instead of the stale register-file value.

Load use dependence

R3 becomes 12; one stall; the earlier store commits once.

Cycle	IF PC	ID	EX	MEM	WB	Stall	Flush	Fwd A/B	Writes at edge
1	00	--	--	--	--	0	0	-/-	-
2	01	00	--	--	--	0	0	-/-	-
3	02	01	00	--	--	0	0	-/-	-
4	03	02	01	00	--	0	0	-/-	-
5	04	03	02	01	00	0	0	EM/WB	R2=10

Cycle	IF PC	ID	EX	MEM	WB	Stall	Flush	Fwd A/B	Writes at edge
6	05	04	03	02	01	0	0	-/-	R0=07, M[10]=07
7	06	05	04	03	02	1	0	-/RF	-
8	06	05	--	04	03	0	0	-/-	R3=05
9	07	06	05	--	04	0	0	RF/WB	R1=07
10	08	07	06	05	--	0	0	-/-	-
11	08	--	--	06	05	0	0	-/-	R3=0C
12	08	--	--	--	06	0	0	-/-	-

A stage address means valid=1; -- means valid=0. IF PC is the fetch address, including when held or halted. EM=EX/MEM forwarding; WB=MEM/WB forwarding; RF=latched register operand; -=unused or invalid. Forwarding labels are derived from the sampled comparator/control inputs; complete raw pins are in the matching CSV. MEM/WB instruction identities follow the previous EX entries because these banks always advance.

Test program and exact stimulus: tests/8_2_load_use.asm and tests/8_2_load_use.stimulus.txt. WB and store entries describe the architectural writes at the following edge. Reset and trailing duplicate show snapshots are omitted from this compact table.

Youngest writer to store

R1 changes 10 to 13 to 9; the store writes 9 at RAM address 12 hex.

Cycle	IF PC	ID	EX	MEM	WB	Stall	Flush	Fwd A/B	Writes at edge
1	00	--	--	--	--	0	0	-/-	-
2	01	00	--	--	--	0	0	-/-	-
3	02	01	00	--	--	0	0	-/-	-
4	03	02	01	00	--	0	0	-/-	-
5	04	03	02	01	00	0	0	-/-	R1=0A
6	05	04	03	02	01	0	0	-/-	R2=03
7	06	05	04	03	02	0	0	RF/RF	R3=04
8	07	06	05	04	03	0	0	EM/RF	R4=12
9	08	07	06	05	04	0	0	EM/RF	R1=0D
10	09	08	07	06	05	0	0	-/RF	R1=09, M[12]=09
11	0A	09	08	07	06	0	0	-/-	-
12	0A	--	--	08	07	0	0	-/-	R5=09
13	0A	--	--	--	08	0	0	-/-	-

A stage address means valid=1; -- means valid=0. IF PC is the fetch address, including when held or halted. EM=EX/MEM forwarding; WB=MEM/WB forwarding; RF=latched register operand; -=unused or invalid. Forwarding labels are derived from the sampled comparator/control inputs; complete raw pins are in the matching CSV. MEM/WB instruction identities follow the previous EX entries because these banks always advance.

Test program and exact stimulus: tests/8_3_youngest.asm and tests/8_3_youngest.stimulus.txt. WB and store entries describe the architectural writes at the following edge. Reset and trailing duplicate show snapshots are omitted from this compact table.

Taken branch kills actuator store

Wrong-path ST and ADD are invalidated; no actuator store commits; R3 becomes 85.

Cycle	IF PC	ID	EX	MEM	WB	Stall	Flush	Fwd A/B	Writes at edge
1	00	--	--	--	--	0	0	-/-	-
2	01	00	--	--	--	0	0	-/-	-
3	02	01	00	--	--	0	0	-/-	-
4	03	02	01	00	--	0	0	-/-	-
5	04	03	02	01	00	0	0	-/-	R1=05
6	05	04	03	02	01	0	0	RF/RF	R2=01
7	06	05	04	03	02	0	1	-/EM	R4=F4
8	07	--	--	04	03	0	0	-/-	R1=00
9	08	07	--	--	04	0	0	-/-	-
10	09	08	07	--	--	0	0	-/-	-
11	0A	09	08	07	--	0	0	-/-	-
12	0A	--	--	08	07	0	0	-/-	R3=55
13	0A	--	--	--	08	0	0	-/-	-

A stage address means valid=1; -- means valid=0. IF PC is the fetch address, including when held or halted. EM=EX/MEM forwarding; WB=MEM/WB forwarding; RF=latched register operand; -=unused or invalid. Forwarding labels are derived from the sampled comparator/control inputs; complete raw pins are in the matching CSV. MEM/WB instruction identities follow the previous EX entries because these banks always advance.

Test program and exact stimulus: tests/8_4_flush.asm and tests/8_4_flush.stimulus.txt. WB and store entries describe the architectural writes at the following edge. Reset and trailing duplicate show snapshots are omitted from this compact table.